

CANDIDATE
NAME

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PDG

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PDG
INDEX NUMBER

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BIOLOGY

9648/02

Paper 2 Core Paper

18 September 2015
Friday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
Section A	80
1	
2	
3	
4	
5	
6	
7	
8	
Section B	20
9 / 10	
Total	100

Section A

Answer **all** the questions in this section.

- 1 The synthesis of collagen is shown in Fig 1.1.

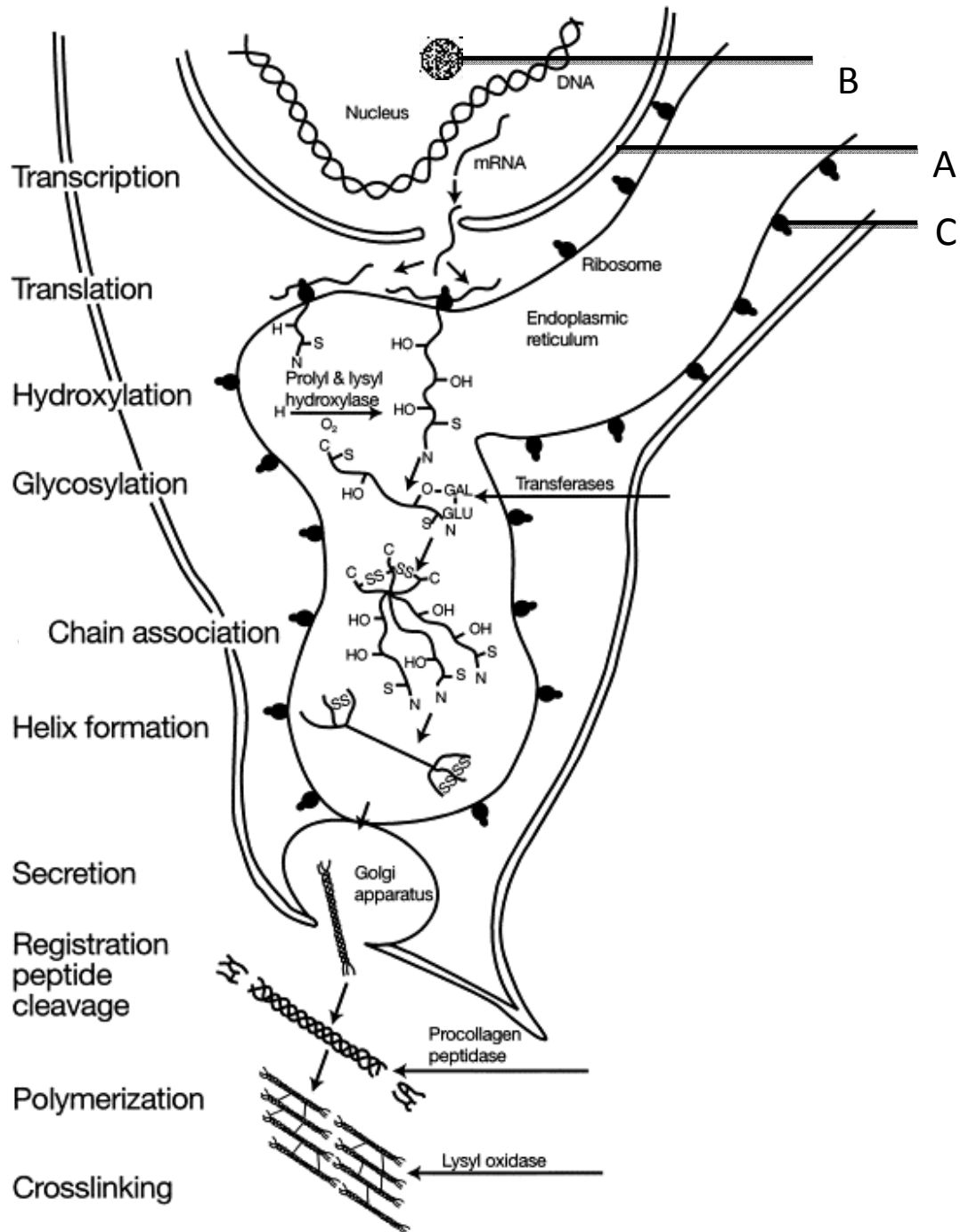


Fig. 1.1

(a)	(i)	State two reasons how A is important to the synthesis of collagen.	
		• A is nuclear envelope with nuclear pores; (Any 1) • Allows mRNA to leave to be translated to collagen by ribosomes in cytosol; • Allows tRNA made to leave to attach to amino acid for collagen synthesis; • Allows RNA polymerase, RNA nucleotides and ATP to enter to make mRNA/ tRNA for synthesis of collagen. AND • A is nuclear envelope which is continuous with RER; • RER has ribosomes which result in synthesis of tropocollagen; Award one mark for each importance.	[2]
	(ii)	Explain how the functions of the two organelles, B and C , are linked.	
		• B is nucleolus and C is ribosome; • B synthesizes rRNA; (Any 2 below) • rRNA makes up the structure of ribosome; • rRNA contributes to the tRNA and mRNA binding sites of ribosome; • rRNA contributes to peptidyl transferase catalytic activity of ribosome; •	[4]
(b)		Suggest how modification such as hydroxylation in the rough endoplasmic reticulum results in collagen having a high tensile strength.	[2]
		• Result in the addition of <u>hydroxyl group/ polar groups</u> • that form <u>hydrogen bonds</u>; • Chain association to form <u>triple helix/ three polypeptides</u> strengthens the tropocollagen;	

(c) A molecule of messenger RNA (mRNA) was produced during the transcription of the collagen gene.

Part of the template sequence of DNA was ATGC. Fig. 1.2 shows the part of the molecule of messenger RNA corresponding to that sequence of four bases.

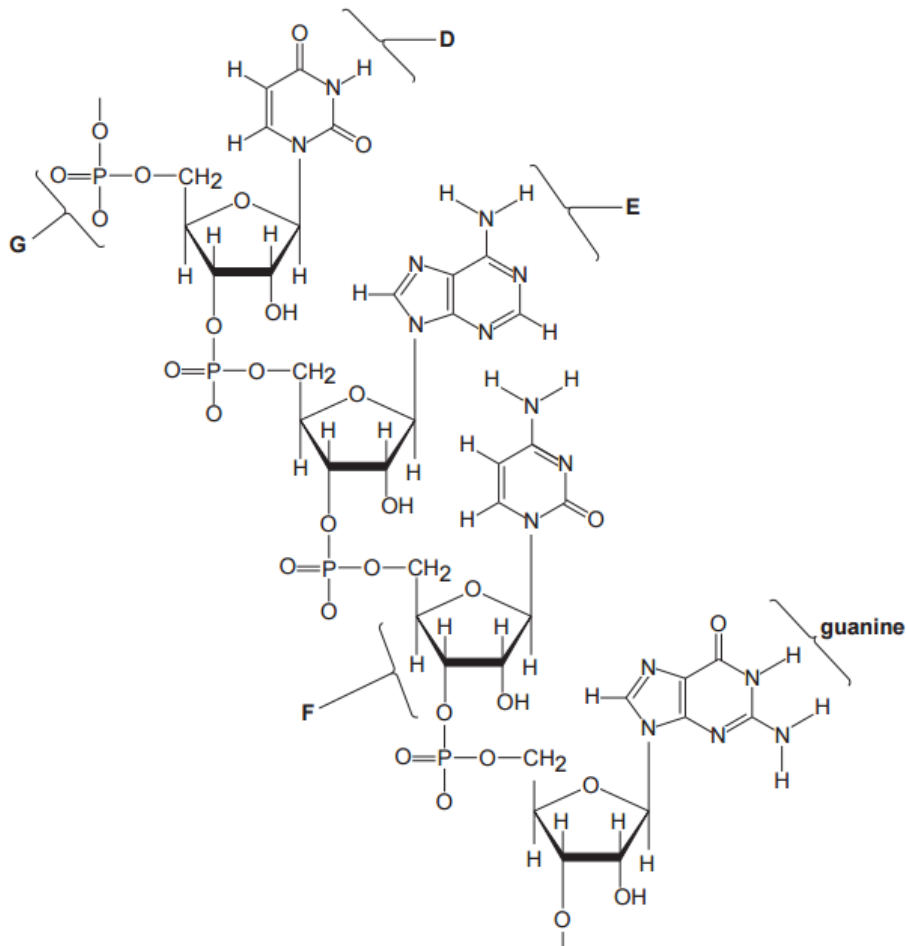


Fig. 1.2

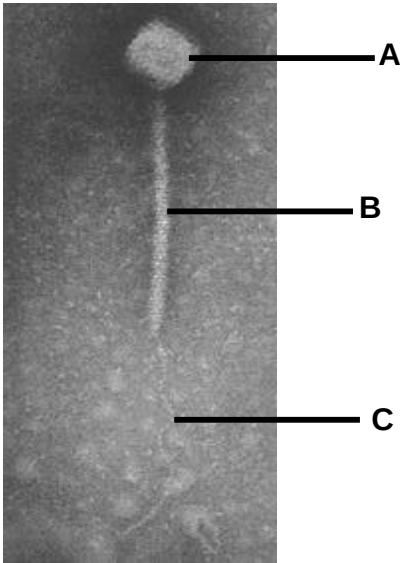
Name the parts of the mRNA molecule shown in Fig. 1.2 labelled D, E, F and G.

[2]

Answer:

D – uracil ; E – adenine ; F – ribose sugar (reject 5C sugar); G – phosphate group

[Total: 10 marks]

2	Fig. 2.1 shows a virus found in a sample of water.  Fig. 2.1	
	(a) State the name of structures A – C.	[3]
	A: capsid/ capsid head/ icosahedral capsid head; B: tail sheath C: tail fiber (reject: tail fibres)	
	(b) Lambda virus undergoes lysogenic pathway after infecting its host cell. Distinguish between the lytic and lysogenic pathway.	[2]
	The following passage is an extract taken from an online article: “For the new study, the researchers developed a particle that works in a similar way - it targets and kills specific bacteria, without causing the cells to burst and release their toxins. They call the particles "phagemids" because they infect the target bacteria with plasmids - small DNA molecules that can copy themselves inside cells. Using synthetic biology, the team engineered the plasmids to express proteins and peptides - short-chain amino acids - that are toxic to the bacterial host cell. The toxins are designed to disrupt key cell processes such as replication, with the effect that the bacterial cell dies without bursting. The team systematically tested a variety of peptides and toxins and showed how, when some are combined in the phagemids, they kill the great majority of bacterial cells within a culture. The method they have developed is highly targeted.”	

	- Medical News Today, 1 st July 2015 http://www.medicalnewstoday.com/articles/296127.php		
	(c)	(i)	Suggest how these toxins mentioned in the passage may work. [2]
			- Competitive/ non-competitive inhibitors that binds to active site/site other than active site of DNA polymerase/ RNA polymerase; - DNA polymerase not able to bind to DNA for replication to take place/ RNA polymerase not able to bind to DNA for transcription to take place to produce essential proteins; - Any AVP (e.g. bind to RNA polymerase DNA binding site/ enzyme that targets DNA polymerase for degradation)
		(ii)	Explain why “the method described is highly targeted”. [1]
			- Phage has specific shape on their tail fibre(s) that are complementary to specific receptors; - Only able to infect/ work on a specific (strain of) bacteria;
		(iii)	Give one advantage and one disadvantage of using phagemids to kill infectious bacteria. [2]
			[Total: 10 marks]

3 Iron ion concentration can be kept constant by transferrin membrane receptor proteins.

At high intracellular iron ion concentration, transferrin receptor proteins are inactive. At low intracellular iron ion concentration, transferrin receptor proteins will transfer and increase iron ions into the cytoplasm.

Production of transferrin receptors are regulated at the mRNA level by active IRE-binding proteins (IRPs) which bind to an iron response element (IRE) that forms a hairpin loop on the 3' untranslated region (3' UTR) of transferrin mRNA as shown in Fig. 3.1.

Transferrin receptor proteins

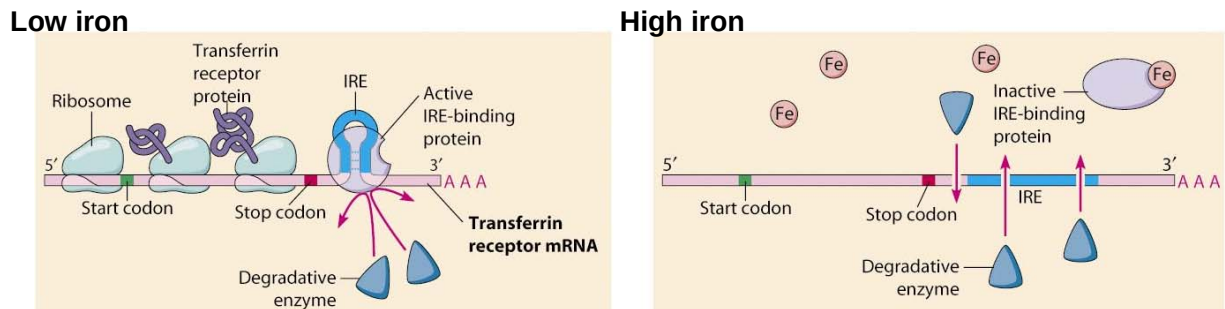


Fig. 3.1

(a) With reference to the structure of transferrin receptor mRNA in Fig. 3.1,

(i) explain how IRE can bind to the active IRP;

Answer:

- IRE folds back on itself via complementary base-pairing/ reference to H bonds;
- Forms 3D complementary shape to IRP binding site;

[2]

(ii) describe how expression of the transferrin receptor genes may be controlled at the **translational** level when iron ion concentration is **high**.

Answer:

- Ref to how iron prevents IRP binding from binding to IRE;
- No hairpin loop;
- (2 marks below compulsory points)
- Degradative enzyme degrades transferrin receptor mRNA at 3' end;
- mRNA not translated by ribosome;

[3]

Cancer has always been associated with telomerase gene expression.

Human telomerase (hTERT) gene was expressed in a cancer cell, and the results are shown in Fig. 3.2.

	Tumor Volume (Mean, in mm ³)				
	10 days	13 days	17 days	20 days	24 days
Control	0	0	4.3	5.05	17.87
hTERT	38.07	115.7	146.29	228.98	319.56

	Tumor Growth (mm ³ /day)	Final Tumor Weight (Mean, in mg)
Control	1.19	Undetermined
hTERT	19.25	342

Fig. 3.2

(b)	With reference to Fig. 3.2,	
(i)	Describe the relationship between hTERT gene expression and cancer.	[2]
	Answer: • Reference to faster rate of tumour growth <u>and</u> <u>processed</u> data; • Reference to increased expression leads to bigger volume <u>and</u> data;	
(ii)	Explain your results.	
	Answer: • hTERT expresses telomerase which lengthen the telomeres of cancer cell DNA; • Telomere not shortened to critical length to cause apoptosis/ senescence; • Able to divide by mitosis for indefinite period of time/ uncontrolled cell division;	[3]

4	The <i>H-ras</i> oncogene codes for the <i>ras</i> protein, which is type of G-protein. Fig. 4.1 is part of the base sequence of a proto-oncogene and of the <i>H-ras</i> oncogene, which arises from the proto-oncogene by mutation. The site of this mutation is where GTPase enzyme of <i>ras</i> protein is located. Fig. 4.1 also shows part of the primary structure of their two encoded proteins. <table><tr><td colspan="14">proto-oncogene</td></tr><tr><td>ATG</td><td>ACG</td><td>GAA</td><td>TAT</td><td>AAG</td><td>CTG</td><td>GTG</td><td>GTG</td><td>GGC</td><td>GCC</td><td>GGC</td><td>GGT</td><td>GTG</td><td>GGC</td></tr><tr><td>met</td><td>thr</td><td>glu</td><td>tyr</td><td>lys</td><td>leu</td><td>val</td><td>val</td><td>gly</td><td>ala</td><td>gly</td><td>gly</td><td>val</td><td>gly</td></tr><tr><td colspan="14">oncogene</td></tr><tr><td>ATG</td><td>ACG</td><td>GAA</td><td>TAT</td><td>AAG</td><td>CTG</td><td>GTG</td><td>GTG</td><td>GGC</td><td>GCC</td><td>GTC</td><td>GGT</td><td>GTG</td><td>GGC</td></tr><tr><td>met</td><td>thr</td><td>glu</td><td>tyr</td><td>lys</td><td>leu</td><td>val</td><td>val</td><td>gly</td><td>ala</td><td>val</td><td>gly</td><td>val</td><td>gly</td></tr></table> Fig. 4.1	proto-oncogene														ATG	ACG	GAA	TAT	AAG	CTG	GTG	GTG	GGC	GCC	GGC	GGT	GTG	GGC	met	thr	glu	tyr	lys	leu	val	val	gly	ala	gly	gly	val	gly	oncogene														ATG	ACG	GAA	TAT	AAG	CTG	GTG	GTG	GGC	GCC	GTC	GGT	GTG	GGC	met	thr	glu	tyr	lys	leu	val	val	gly	ala	val	gly	val	gly	
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(a)	State the first three codons of mRNA of <i>ras</i> proto-oncogene.	[1]																																																																																				
	AUG ACG GAA																																																																																					
(b)	Explain how this mutation may change the structure and function of <i>ras</i> protein.																																																																																					
	• Base substitution mutation of G to T; • At 11th DNA base triplet; • Cause mRNA codon to change from GGC to GUC; • results in the formation of valine instead of glycine in the <i>ras</i> protein; • Valine is hydrophobic; • Ref to R-group interactions affected/ hydrophobic R group of valine forms hydrophobic interaction with other amino acid of oncogene polypeptide chain; • causing the polypeptide to undergo folding to different 3D tertiary structural conformation; • GTPase enzyme non-functional; • cannot hydrolysed GTP to GDP; • <i>ras</i> protein becomes hyperactive;	[4]																																																																																				

Fig. 4.2 shows the cell cycle-stimulating pathway that involves *ras* protein.

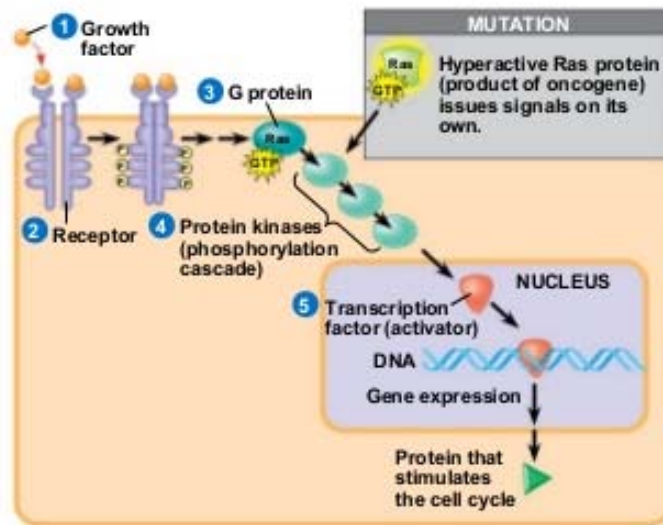


Fig. 4.2

(c) Using Fig. 4.2, describe how a mutated *ras* protein can stimulate cell cycle.

- **hyperactive *ras* protein is activated;**
- even in **absence** of signal/ growth factor;
- has **GTP binds to it permanently/ GTPase cannot hydrolysed GTP to GDP;**
- activates **protein kinases;**
- in **phosphorylation cascade;**
- activate **transcription factor;**
- **TF binds to promoter with RNA polymerase;**
- trigger **gene expression of proteins** that stimulate cell cycle;

[3]

(d) It was found that a simple gene amplification of *ras* proto-oncogene is less likely to be oncogenic. Suggest why.

- **more normal *ras* proteins produced;**
- **capable of on/off regulation/ capable to bind to GDP or GTPase still functional**

[2]

[Total: 10 marks]

5	(a)	Distinguish between the inheritance of multiple genes and multiple alleles.	[2]
		In cattle, coats may be solid white, solid black, or black-and-white-spotted. When true breeding solid whites are mated with true-breeding solid blacks, the F1 generation consists of all solid white individuals. Matings among the F1 generation have resulted in the following ratio: 23 solid white 6 black-and-white spotted 3 solid black	
	(b) (i)	Is coat colour in cattle an example of multigenic inheritance or an example of multiple alleles at one gene locus?	[1]
	(ii)	Explain the mode of inheritance observed.	[2]
	(c)	Using a genetic diagram, predict the genotypes and phenotypes of a cross between a black-and-white spotted cow with a black parent, and a heterozygous solid white cow.	[3]
	(d)	Explain why it would be useful to carry out a chi-squared test on these results. No calculations are required to answer this question.	[2]
			[Total: 10 marks]

ANSWERS

(a) Distinguish between multigenic inheritance and multiple alleles. [2]

MULTIGENIC INHERITANCE	MULTIPLE ALLELES
Involves more than one gene locus; [R: chromosome locus!]	Involves only one gene locus;
May not have more than two alleles present at each gene locus;	More than two alleles present at one gene locus;
Interaction of various gene products gives rise to certain traits / characteristics; that lead to an overall phenotype	Interaction of allelic products gives rise to certain trait / characteristic; that will affect the overall phenotype
Additive effects of multiple genes at various gene loci;	No additive effect as only one gene locus is involved;
Results in continuous distribution;	Results in discontinuous distribution;

(b)(i) Is coat colour in cattle an example of multigenic inheritance or an example of multiple alleles at one gene locus? [1]

Multigenic inheritance;;

(ii) Explain the mode of inheritance observed. [2]

Dominant; epistasis;

The presence of the dominant allele at the white locus; that is epistatic; to & prevents the expression of any alleles at the black locus; giving rise a ratio of 12:3:1;

(d) Using a genetic diagram, predict the genotypes and phenotypes of a cross between a black-and-white spotted cow with a black parent, and a heterozygous solid white cow. [5]

Symbols:

A = Dominant allele that inhibits pigment production **a** = Recessive allele that allows pigment production
B = Dominant allele for black-and-white spotted pigment **b** = Recessive allele for black pigment

Parental genotype: $aaBb$; x $AaBb$;

Parental phenotype: Black-and-white spotted Solid white

Gametes: aB ab ; AB Ab aB ab ;

Punnett Square:

Female gametes	Male gametes			
	aB	Ab	aB	ab
aB	$AaBB$	$AaBb$	$aaBB$	$aaBb$
ab	$AaBb$	$Aabb$	$aaBb$	$aabb$

Offspring genotypic ratio: $3 A_B_ : 1 A_bb : 3 aaB_ : 1 aabb$;

Offspring phenotypic ratio: $4 \text{ solid white} : 3 \text{ black-and-white spotted} : 1 \text{ solid black}$;

- (d) To determine if the difference between the observed and expected results was significant or due to chance;
 Difference is significant if the probability is greater than 0.05/ @P=0.05, calculated χ^2 value is greater than the expected χ^2 value;
 (vice versa)

EC: The purpose of the chi-squared test to determine if the difference between observed and expected results was significant, or due to chance, was well known, but further detail about probabilities or differences from expected ratios was usually omitted.

- 6 During the process of glycolysis, glucose is converted by a series of steps into two molecules of pyruvate. Fig. 6.1 outlines glycolysis.

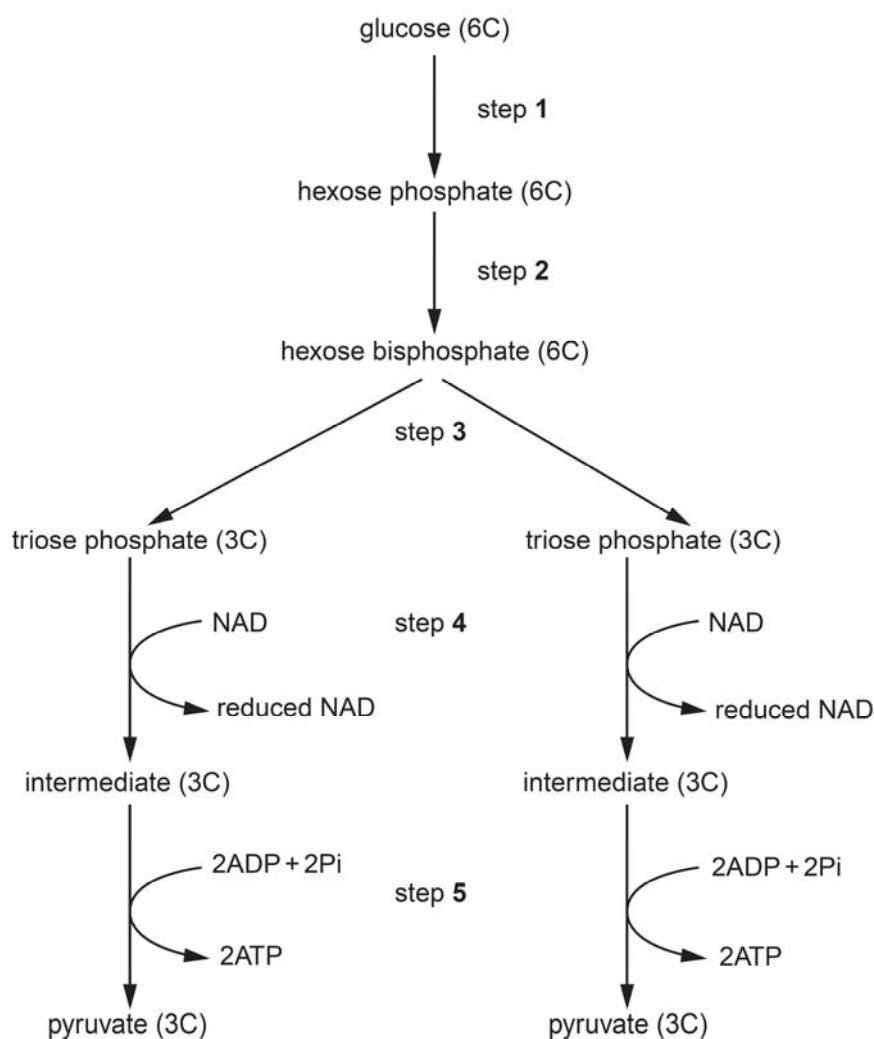


Fig. 6.1

- (a) With reference to Fig. 6.1, state the reactions occurring at:

- (i) steps 1 and 2
- (i) step 3
- (ii) step 4.

[2]

Phosphorylation, Lysis/ hydrolysis, Dehydrogenation or Oxidation

- (b) List **three** ways in which the synthesis of ATP in cytoplasm is different from the synthesis of ATP in mitochondria.

Features	Substrate level phosphorylation	Oxidative phosphorylation
ATP synthesis	Synthesis of ATP from direct enzymatic transfer of phosphate group from substrate to ADP	Synthesis of ATP occurs as electrons are transferred from NADH or FADH ₂ to oxygen down ETC

[3]

	Source of energy for ATP synthesis	Transfer of phosphate group from substrate releases sufficient energy to drive the synthesis of ATP	Energy required for ATP synthesis is coupled to movement of protons down a proton gradient (proton motive force)	
	Source of phosphate	From phosphorylated organic molecule (substrate) to ADP	Phosphorylation of ADP from free inorganic phosphate	
	Location	In the cytosol (glycolysis) and matrix of mitochondrion (Krebs cycle)	On the inner mitochondrion membrane	
	Involvement of ETC	No	Yes	
	Involvement of redox	No	Series of redox	
	Use of coenzymes	Does not involve NADH or FADH ₂	NADH or FADH ₂ transfer electron to the electron carriers and are regenerated to NAD and FAD.	
	Net number of ATP per glucose molecule oxidized	4	34	
(c)	Explain how ATP is able to transfer energy in cells.			
	• Synthesized from ADP and Pi; • Soluble molecule • Diffuses rapidly/ transported easily; • On hydrolysis of third phosphate; • Energy released/ 30.5 kJ(mol⁻¹) • Ref (idea) intermediary (between energy yielding and energy requiring reaction);			
(d)	Fig. 6.2 outlines anaerobic respiration in yeast cells. Complete Fig. 6.2 by writing in the missing compounds.			
	glucose ↓ pyruvate → 2H → 			

- 7 The evolution of haemoglobin molecules has been studied extensively by comparing the amino acid sequences in both myoglobin and haemoglobin. About 500 million years ago, a gene duplication event occurred and one copy became the present day myoglobin and the other evolved into an oxygen-transport protein that gave rise to the present day haemoglobin (Fig. 7.1).

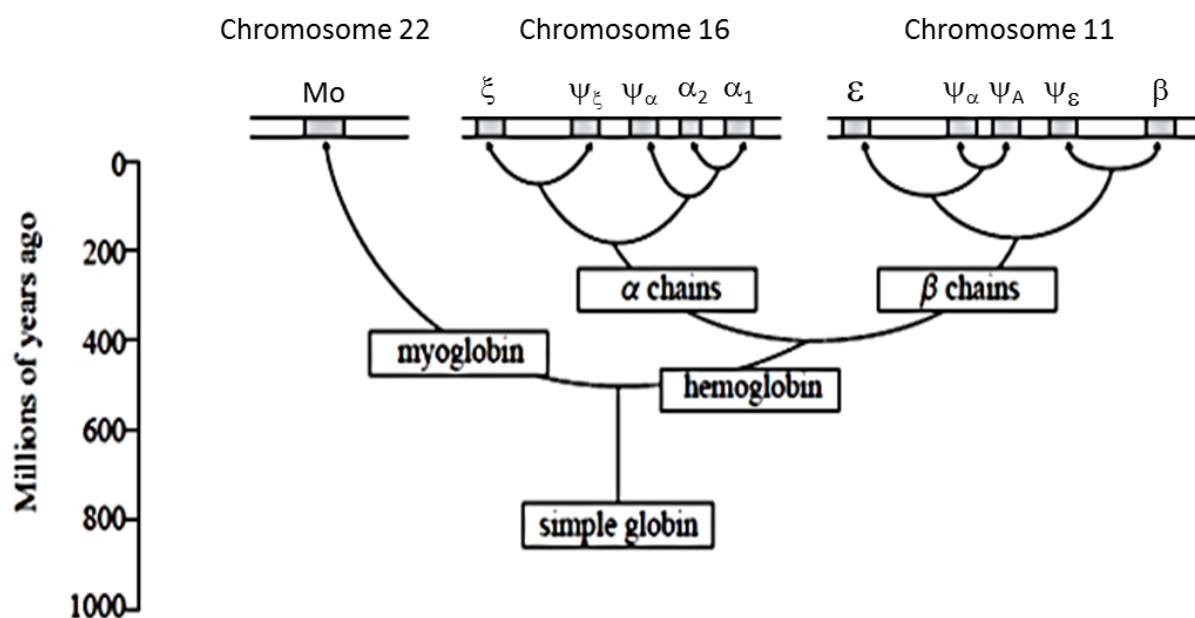


Fig. 7.1

- (a) State how many years ago the simple globin molecule diverged into myoglobin and haemoglobin.

- 500 million years ago

[1]

- (b) Suggest why changes observed in the sequence of amino acids may lead to an under-estimation of the actual number of mutations.

[2]

- A mutation may result in the same amino acid being produced due to genetic code being degenerate
- Some mutations may have occurred in the non-coding DNA region / introns

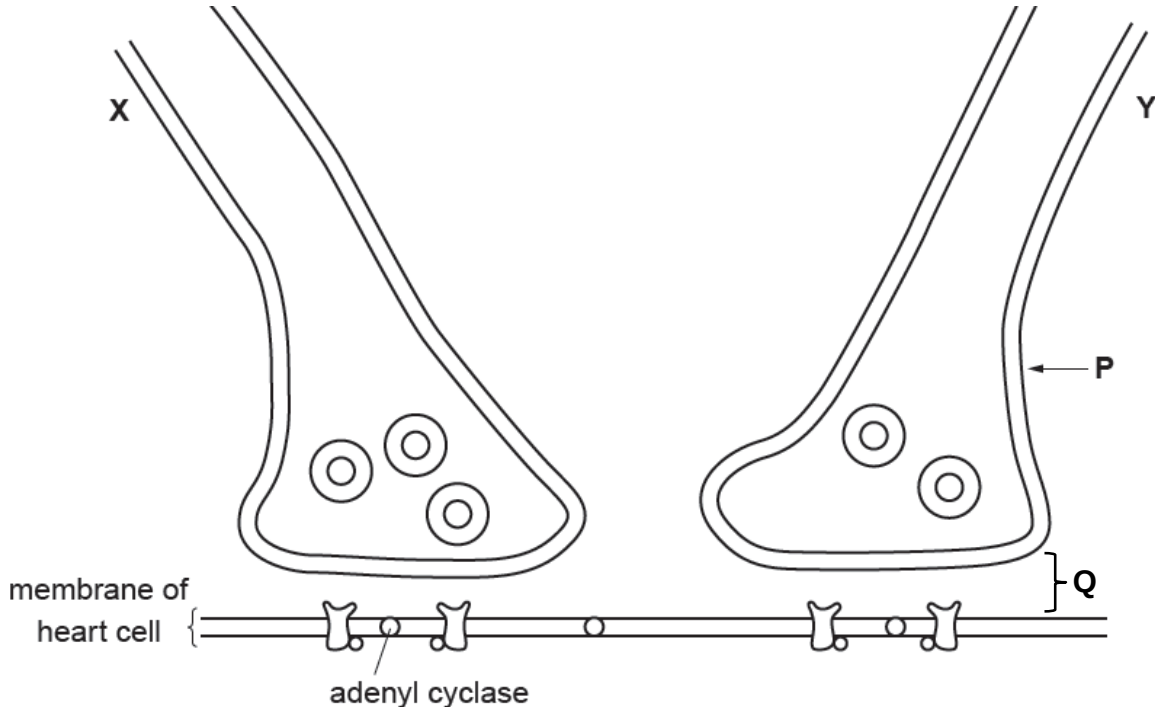
- (c) Explain how variations in the haemoglobin gene can be used as a molecular clock.

[3]

- The mutation in the gene coding for haemoglobin occurred at a constant rate through time
- The number of nucleotide differences can be used as a gauge to estimate when two species diverged from a common ancestor / **OWTTE**
- The lesser the number of nucleotide difference, the more closely related the two species are [**A: reverse argument**]

	(d) Haemoglobin transports oxygen in red blood cells of all vertebrates such as fishes. The common name of <i>Gambusia affinis</i> is mosquitofish. Explain why the scientific name is better than the common names.													
	1. It is <u>unambiguous</u> and precise with one unique name for one species; 2. unlike the common name which may be used to refer to two species or multiple names referring to one species; 3. The <u>binomial nomenclature</u> is a <u>systematic naming</u> system that allows one to categorise/group similar organisms together involving <u>both the species name and genus name</u> (evidence); 4. while the common names do not <u>reveal their position in the animal kingdom</u> nor attempts to categorise them;	[2]												
	(e) Scientific names reflect the classification of organisms. Describe how classification differs from phylogeny.													
	<table border="1"> <thead> <tr> <th data-bbox="236 797 435 831">Feature</th><th data-bbox="435 797 917 831">Classification</th><th data-bbox="917 797 1380 831">Phylogeny</th></tr> </thead> <tbody> <tr> <td data-bbox="236 831 435 1384">Basis of grouping</td><td data-bbox="435 831 917 1384"> Grouping based on similarities in characteristics and does not consider evolutionary relationships. Organisms with many similarities in observable characteristics are grouped together. Grouping does not consider descent with modification </td><td data-bbox="917 831 1380 1384"> Grouping based on examining mainly molecular homology such as DNA sequence to derive ancestry and evolutionary history / relationships. The more homologies shared between two species, the more recent they diverge from a common ancestor. Grouping considers the passing of genes from ancestors to their descendants and descent with modification. </td></tr> <tr> <td data-bbox="236 1384 435 1910">Format of organisation</td><td data-bbox="435 1384 917 1910"> Organisation of organisms into a hierarchy of increasingly broad categories. Species that are closely related are assigned to the same genus and related genera are grouped into a single family. Families are grouped into orders, orders into classes, classes into phyla (singular phylum) and phyla into kingdoms and more recently, kingdoms into domains A binomial naming system is used. </td><td data-bbox="917 1384 1380 1910"> Organisation is in the form of phylogenetic tree Phylogenetic tree is a branching diagram which shows descent/ divergence of different species from a common ancestor. </td></tr> <tr> <td data-bbox="236 1910 435 1977">Natural / man-made</td><td data-bbox="435 1910 917 1977"> Classification is human effort to give order to data via Linnaean </td><td data-bbox="917 1910 1380 1977"> Phylogeny reflects natural relationship among organisms. </td></tr> </tbody> </table>	Feature	Classification	Phylogeny	Basis of grouping	Grouping based on similarities in characteristics and does not consider evolutionary relationships. Organisms with many similarities in observable characteristics are grouped together. Grouping does not consider descent with modification	Grouping based on examining mainly molecular homology such as DNA sequence to derive ancestry and evolutionary history / relationships. The more homologies shared between two species, the more recent they diverge from a common ancestor. Grouping considers the passing of genes from ancestors to their descendants and descent with modification.	Format of organisation	Organisation of organisms into a hierarchy of increasingly broad categories. Species that are closely related are assigned to the same genus and related genera are grouped into a single family. Families are grouped into orders, orders into classes, classes into phyla (singular phylum) and phyla into kingdoms and more recently, kingdoms into domains A binomial naming system is used.	Organisation is in the form of phylogenetic tree Phylogenetic tree is a branching diagram which shows descent/ divergence of different species from a common ancestor.	Natural / man-made	Classification is human effort to give order to data via Linnaean	Phylogeny reflects natural relationship among organisms.	[2]
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Natural / man-made	Classification is human effort to give order to data via Linnaean	Phylogeny reflects natural relationship among organisms.												

		organisation	binomial nomenclature and hierarchy to ensure that organisms can be universally identified by their scientific name.		
		[Total: 10 marks]			

8	Fig. 8.1 shows endings of neurones X and Y in the heart. Stimulation of Y leads to an increase in the concentration of the secondary messenger cyclic AMP (cAMP), in the cytoplasm of the heart cell and the stimulation of X leads to a decrease.  Fig. 8.1	
(a)	Describe the events that follow the arrival of an impulse at P on neurone Y that leads to an increase in cAMP.	[5]
	• Voltage-gated calcium channel opens leading to an influx of calcium ions into the neurone; • Calcium ions cause movement of neurotransmitter vesicles towards post-synaptic membrane (along microtubules); • Vesicles fuse with post-synaptic membrane and exit via exocytosis and diffuse across the synaptic cleft; • Neurotransmitter binds to G protein coupled receptor on the membrane of heart cell and activates G protein; • Adenyl cyclase becomes activated and converts ATP to cAMP;	
(b)	The intake of drugs such as cocaine alters nerve impulse transmission between neurones, causing an increase in heart rate. Suggest how this occurs.	[2]
	• Prevents reuptake of neurotransmitters/ break down of neurotransmitters/ binds neurotransmitters permanently to receptors; • Causing constant depolarization to occur at post-synaptic membrane that initiates action potential at a <u>higher frequency</u> down the post-synaptic neurone to cause heart to pump faster; OR answers blockage of IPSPs leading to higher frequency of action potential	

(c)	Distinguish between the impulse that arrives at P and the signal that crosses Q.			[3]
	Feature	Neurone	Synapse	
	Nature of message	Propagation of electrical impulse as action potential	Chemical transmission as neurotransmitters	
	Neurones involved	AP travels along <u>one</u> neurone	2 adjacent neurones	
	Excitatory/Inhibitory	Excitatory	Can be Excitatory OR Inhibitory	
	Time taken to generate AP	Faster (consists of time is taken for movement of ions across axon)	Synaptic delay – time for release of neurotransmitters & diffusion	
	Response	All-or-nothing law → all AP are propagated	Graded potential are not propagated. Summation needs to exceed threshold potential to generate a action potential (at post synaptic membrane)	
	Recovery of resting potential	Recovery to RP by repolarisation and hyperpolarisation (opening of voltage gated K^+ channel thus efflux of K^+)	Recovery to RP by enzymatic action to hydrolyse neurotransmitter/ to reabsorb back the neurotransmitter e.g. acetylcholinesterase which hydrolyses acetylcholine	
	Refractory period	Absolute & Relative	None but synapses can experience synapse fatigue (supply of neurotransmitter exhausted)	
	Speed of transmission	Affected by presence/absence of myelin sheath, diameter of axon, temperature	Drugs ie Nicotine (similar structure to Ach) and Botulinum toxin (prevent release of Ach at presynaptic membrane)	
	Types of gates	Voltage gated Na^+ and K^+ channels	Presynaptic membrane – voltage gated Ca^{2+} channels Post-synaptic membrane – Ligand-gated $Na^+/K^+/Cl^-$ channels	
	Use of ATP	For Na^+/K^+ pumps to maintaining resting potential	For Na^+/K^+ pumps to maintaining resting potential for movement of synaptic vesicles towards pre-synaptic membrane use of Ca^{2+} pumps to maintain low levels of intracellular Ca^{2+} for its influx to trigger vesicles movement	

Section B

Answer **EITHER 9 or 10.**

Write your answers on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in section (a), (b) etc., as indicated in the question.

9	(a)	Using <i>lac</i> operon as an example, describe how prokaryotes such as <i>E.coli</i> is able to respond to changes in the environment.			[7]
		• <i>Lac</i> operon consists of promoter, operator and structural genes; • <i>lacZ</i> gene encodes for beta-galactosidase, an enzyme which hydrolyzes lactose; • <i>LacY</i> gene encodes for galactoside permease to actively transport lactose into the cell; • Genes coding for protein involved in metabolizing lactose are grouped together under a single promoter; When lactose is present • Enters through galactoside permease on cell membrane; • Isomerizes to allolactose by beta-galactosidase; • Allolactose binds to repressor protein, causing repressor to change its shape and becomes inactive; • DNA binding site of repressor is no longer complementary to operator and no longer binds; • RNA polymerase can bind to promoter and transcribe structural genes; • <i>LacZ</i>, <i>Y</i> and <i>A</i> transcribed as a single polycistronic mRNA enabling required proteins to be translated together • respond quickly to the presence of lactose in the environment; OR reverse scenario of lactose is absent When both glucose and lactose are present, • cAMP levels will be low and CAP remains inactive; • CAP does not bind to CAP binding site; • RNA polymerase affinity to promoter is low and rate of transcription of <i>lac</i> operon is low; • Allows bacteria to use glucose as respiratory substrate first before lactose • This saves energy and resources; [each point ½ mark]			
	(b)	Describe the cellular responses that glucagon initiates in order to regulate blood glucose concentration.			[6]
		• Glucagon binds to extracellular ligand-binding site of GPCR • (cytoplasmic domain) of GPCR undergoes conformational change → receptor becomes activated. • Cause bound G-protein to undergo conformational change, releasing its bound GDP, which is displaced by GTP → activated. • Activated G-protein dissociates from receptor • and activates adenylyl cyclase			

- which catalyses the conversion of **ATP to cAMP**.
- cAMP (2nd messenger)
- then activates protein kinase A.
- Activated protein kinase A then phosphorylates other proteins (kinases) in a phosphorylation cascade.
- Glycogen phosphorylase catalyses the breakdown of glycogen to glucose (**increased glycogenolysis** in liver and muscle cells).
- Other cellular response:
 - o **Increase** activity or synthesis of enzymes involved in **gluconeogenesis** (synthesis of glucose from non-carbohydrate sources) → increased rate of gluconeogenesis.
 - o Breakdown of lipids
- After the specific response has been carried out, the signal is terminated.
- Glucagon is released/dissociates from the glucagon GPCR.
- GTP on G-protein is hydrolysed to GDP by intrinsic GTPase enzyme in the G-protein
- cAMP is degraded to AMP by phosphodiesterase.

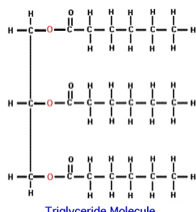
(c) Compare triglyceride and starch as a storage molecule.

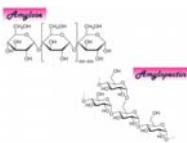
[7]

Similarities

- Both have high percentage of hydrocarbons [not just hydrogen or carbon atoms] and hence stores energy;;
- Both are macromolecules formed through condensation reactions [accept both formation results in loss of water/ production of water; R! polymer as triglyceride is not a polymer];;
- Both are compact [triglyceride consists of non-polar hydrocarbon chains and is dehydrated and can be packed compactly while starch consists of two compact forms – amylose and amylopectin];;
- Both are insoluble and therefore do not affect water potential of cells;;

Feature	Triglyceride	Starch
Basic units	Triglyceride is made up of 1 glycerol and 3 fatty acid chains;	Starch is made up of repeating units of α glucose;
Type of Bonds	bonds linking up the monomers are ester linkages;	bonds linking up the monomers are glycosidic linkages;
Type of biomolecule	Triglyceride is not a polymer;	Starch is a polymer of glucose;
As energy store	Higher/ twice amount of energy stored/ superior long term energy store; (g for g, stores twice the amt of en compared to CHO) Due to greater number of carbon atoms/ more C-H bonds for the same mass;	Lower/ 2X less energy stored; Due to more –OH groups and so less carbon atoms/ less C-H bonds for the same mass;



		 Starch	→ lesser amount of fats needs to be stored for the same amount of energy; A! more compact per unit mass; due to little association of –CH groups with water/ minimal hydration;	→ more starch molecules need to be stored for the same amount of energy; A! less compact per unit mass; due to association of polar – OH groups on surface associating with water;	
					[Total: 20]

	(c)	Discuss the factors that contribute to speciation.	[7]
		1. Speciation is the <u>formation of new species</u> / <u>divergence of existing species into new species</u> 2. <u>Spontaneous mutations</u> give rise to <u>variation</u> within the population... 3. ...which allow <u>favourable characteristics to be selected for</u> and <u>unfavourable characteristics to be selected against</u>. 4. Speciation is partly due to <u>disruption to gene flow</u> in a population of a particular species. Reduced gene flow as a result of: 5. <u>Behavioural isolation</u>, where in a population in a particular area, some members of the population exhibit variation in behaviour and <u>tended to mate preferentially</u>. 6. <u>Geographical isolation</u>, where in a <u>physical barrier</u> separates members of a population into 2 sub populations, preventing interbreeding. 7. <u>Physiological isolation</u>, various mechanical, physical, and biochemical differences between members of a population become barriers to interbreeding between populations, thus maintaining separate, distinct species 8. Any other isolation, e.g. gametic isolation 9. <u>Genetic drift</u> / <u>Founder's effect</u> / <u>Bottleneck effect</u> <u>changes the allele frequencies</u> within the population by chance. 10. <u>Accumulation of mutations / differences</u> over many generations 11. Over many successive generations, each subpopulation became 2 <u>genetically distinct species</u> which were <u>unable to interbreed</u> and produce fertile viable offspring.	
			[Total:20]